SGT HydroEdge Private Limited

SGT GreenX™

Controlled Hydrogen Fuel Assist for Diesel Generators

Credibility & Technology Validation Dossier

For Investors, Customers & Institutional Partners
March 2026

Up to 80%	**8–15%**	**Up to 35%**	**Up to 41%**	**Up to 43%**
Particulate Matter	**kWh/L improvement**	**Nitrogen Oxides**	**Carbon Monoxide**	**Carbon Dioxide**
PM Reduction	Fuel Savings	NOx Reduction	CO Reduction	CO_2 Reduction

Validated Across 5 DG Platforms | 250 kVA to 1500 kVA | NABL-Accredited Lab Testing

Executive Summary

India operates more than 10 million diesel generators — the backbone of backup and captive power for manufacturing plants, hospitals, telecom towers, data centres, ports, and commercial buildings. These DG sets collectively consume billions of litres of diesel annually, generating millions of tonnes of PM 2.5, NOx, CO, HC, and CO_2. Current CPCB emission norms address new DG sets but leave millions of legacy units — and their CO_2 emissions — entirely unregulated.

SGT has developed GreenX™, a Controlled Hydrogen Fuel Assist (CHFA) retrofit that installs on any diesel generator from 25 kVA to 4,000 kVA without engine modification. By generating hydrogen-oxygen (HHO) gas on-demand through alkaline water electrolysis and introducing it into the engine air intake at precisely controlled flow rates, GreenX™ improves combustion completeness — directly reducing fuel consumption and emissions at source, not through exhaust aftertreatment.

This dossier presents the complete body of evidence behind GreenX™, including:

- NABL-accredited laboratory stack emission tests conducted across five DG platforms (250 to 1500 kVA) at industrial customer sites
- Peer-reviewed research paper by Alok Kumar & Laxmikant Karande (SGT), published and citing 21 international journal studies
- An unprecedented joint testing partnership with ARAI (Automotive Research Association of India), targeting CPCB recognition as an official emission retrofit technology
- A signed OEM collaboration with Kirloskar Oil Engines Ltd (KOEL), India's largest DG set manufacturer
- Selection as a Top 10 Sustainability Company in India by CII-ASCEND, with CII co-funding the KOEL pilot POC
- Commercial validation from two industrial customers with one placing confirmed repeat orders
- A dedicated ESG and Scope 1 value framework demonstrating GreenX™ contribution to Net-Zero and BRSR targets across DG capacities

SGT GreenX™ is not a concept product. It is a field-tested, institutionally validated, commercially deployed solution occupying a unique position: the only Indian company with this level of ARAI, OEM, and NABL-validated evidence for hydrogen fuel assist on diesel generators.

Company & Institutional Standing

About SGT

SGT HydroEdge Private Limited is a bootstrapped climate-tech startup registered in 2023, headquartered in Pune, Maharashtra. The company was founded by Alok Kumar, an industry veteran with deep expertise in automotive, R&D, marine, and logistics sectors.

Founded	2023 | Pune, Maharashtra | Bootstrapped
Team	12-member cross-functional team spanning engineering, R&D, IoT, and commercial functions
Mission	Decarbonisation-as-a-Service (DaaS) — measurable, profitable, ESG-compliant decarbonisation for every engine, fleet, and enterprise
Technology	Patented HHO generators (GreenX™ for DGs, GreenDrive™ for vehicles, GreenMarine™ for ships)
Awards	Energy Leap Award 2025 for Innovation in Green Hydrogen Application (with grant) | CII-ASCEND Top 10 Sustainability Company in India 2025-26

Institutional Recognitions & Partnerships

1. CII-ASCEND — Top 10 Sustainability Company in India

The Confederation of Indian Industry (CII) ASCEND programme is India's premier industry-led cleantech accelerator. In 2025-26, CII-ASCEND selected Saarthi GreenTech from hundreds of applicants as one of the Top 10 Sustainability Companies in India. This recognition came with:

- Co-funding for the KOEL pilot POC demonstration at Kirloskar Oil Engines' Khadki, Pune facility
- Facilitation of CII Measurement & Verification (M&V) methodology at the pilot site
- Permission to publish and disseminate results across CII's industrial network
- Access to Chief Sustainability Officers and ESG leaders of CII member companies

The CII-ASCEND agreement includes formal measurement & verification protocols, site visit facilitation, photography/documentation rights, and publication consent — creating a fully transparent and independently witnessed pilot record.

2. KOEL (Kirloskar Oil Engines Ltd) — OEM Partnership

In an unprecedented move for a startup of SGT's stage, Kirloskar Oil Engines Ltd (KOEL) — India's largest diesel engine and generator OEM — has formally signed a Consent and Undertakings letter to host the GreenX™ pilot demonstration at their Laxmanrao Kirloskar Road, Khadki, Pune facility. The consent letter, dated 18th February 2026, is signed by Dr. Yogesh Aghav, Vice President, Corporate Research & Engineering.

The significance of OEM participation cannot be overstated. KOEL's agreement means:

- An Indian DG OEM is actively testing and validating CHFA technology on their own equipment

- Results from this pilot will carry OEM credibility and can influence KOEL's own product roadmap
- The testing protocol follows CII M&V standards, ensuring independent verification
- KOEL's network reaches thousands of DG operators across India — creating a direct commercialisation pathway post-validation

In our CHFA research paper, a 2023-model KOEL 500 kVA DG set (Engine E, tested at Cilicant Pvt. Ltd., Pune) was already included as one of the five test platforms, producing verified emission reduction data at NABL-accredited Neetal Laboratories.

3. ARAI — Joint Testing Partnership

The Automotive Research Association of India (ARAI) is India's premier government-accredited automotive testing and R&D organisation under the Ministry of Heavy Industries. ARAI accreditation is the prerequisite for any vehicle or engine technology to receive Government of India regulatory approval.

In an unprecedented move for a startup, ARAI has offered to conduct joint testing with SGT's GreenX™ technology — with the explicit goal of establishing it as a recognised emission retrofit technology for DG sets under CPCB norms. This pathway, if successfully completed, would mean:

- GreenX™ becomes the first hydrogen fuel assist technology officially recognised by CPCB for DG emission reduction
- All legacy DG sets in India become eligible to adopt GreenX™ under a government-recognised framework
- The technology can be extended to marine engines, which share similar diesel combustion architectures
- SGT gains a competitive moat that no competitor can easily replicate — regulatory recognition takes years

Our CHFA research paper (Section 6.6) explicitly identifies ARAI's role: 'Institutions such as ARAI are uniquely positioned to establish standardised test protocols for CHFA evaluation, validate repeatability, safety, and long-term durability, support evidence-based policy formulation for combustion efficiency technologies, and facilitate collaboration between technology developers, regulators, and OEMs.'

GreenX™ Technology — How It Works

The Problem: Incomplete Diesel Combustion

No diesel engine achieves 100% combustion efficiency. In a typical DG set operating at 40–70% load (the most common industrial duty cycle), significant quantities of fuel molecules escape combustion and exit as PM, CO, HC, and CO_2 — representing both wasted energy and direct emissions. Post-combustion aftertreatment systems (DPF, SCR, DOC) attempt to clean up this incomplete combustion, but they:

- Add exhaust backpressure, increasing fuel consumption by 2–5%
- Generate N_2O from SCR reactions — a greenhouse gas 265x more potent than CO_2
- Do not reduce CO_2 emissions, which directly track fuel burn
- Require expensive consumables (urea/AdBlue) and frequent maintenance

The GreenX™ Solution: Pre-Combustion Enhancement

GreenX™ is a pre-combustion technology. It addresses the root cause — incomplete combustion — rather than treating symptoms in the exhaust. The system generates HHO (hydrogen-oxygen) gas on demand through alkaline water electrolysis, then injects it into the DG set's air intake manifold at precisely controlled flow rates using adaptive algorithms.

Property	Hydrogen (H_2)	Diesel Vapour
Minimum Ignition Energy	0.02 mJ (12× lower than diesel)	~0.24 mJ
Laminar Flame Speed	~2.91 m/s (7× faster)	~0.35 m/s
Flammability Range in Air	4–75% by volume (ultra-wide)	0.6–7.5% by volume
Adiabatic Flame Temp	~2400 K	~2200 K

The Three-Zone Operating Model

SGT's research paper is the first to systematically define and experimentally validate the three operating zones for hydrogen fuel assist:

- Zone A (Under-dosing): Hydrogen too low — minimal combustion impact, negligible benefit
- Zone B (Optimal Window): Hydrogen at precisely calibrated level — combustion completeness improves, fuel savings and emission reductions achieved. This is GreenX™'s operating target.
- Zone C (Over-dosing): Hydrogen too high — combustion phasing advances excessively, efficiency degrades, NOx can increase

The existence of this bell-shaped efficiency response explains why prior academic literature showed contradictory results — studies inadvertently operated in Zone C and reported degradation, while studies in Zone B reported improvement. SGT's adaptive control algorithm continuously maintains Zone B operation across varying engine loads.

Why Adaptive Control is SGT's Differentiator

GreenX™'s patent-applied design includes adaptive, load-dependent dosing control that accounts for engine RPM, load, injection pressure, and tuning state. This is not a fixed-flow system. The research paper demonstrates that this matters critically:

- Older CPCB 2 engines (2006–2012 vintage) tolerate wider hydrogen windows and show 8–15% kWh/L improvement
- Newer CPCB 4 engines (2020+ vintage) require tighter, lower dosing — optimal window is at 7–8A vs 12–13A for older engines
- The same dosing level that gives the best result on a 2006 Caterpillar would cause efficiency degradation on a 2023 KOEL engine

A fixed-flow system cannot serve both engine generations correctly. GreenX™'s adaptive algorithm is essential — and is a key element of the patent filing.

System Specifications

Parameter	Specification
Applicable DG Range	25 kVA to 4,000 kVA (any manufacturer)
HHO Flow Rate	160 ml/min to 750+ ml/min (adaptive, load-matched)
Input Required	Water + DC electricity from DG canopy supply. No external power source required.
Engine Modification	None. Retrofit to air intake only. No changes to fuel system, injection timing, or ECU.
Hydrogen Storage	Zero storage. HHO generated on-demand. Auto-shutdown on engine stop. Fully safe.
Connectivity	IoT-enabled. Integrates with SGT GreenVision™ Digital Twin platform for real-time monitoring.
Duty Cycle	24×7 continuous industrial operation rated
IP Protection	Patent applied (India). Additional patents under process.
DG Set Compliance	Compatible with CPCB 2, CPCB 4, CPCB 4+ emission norm DG sets

Test Evidence — NABL-Accredited Laboratory Results

All emission testing was conducted by NABL-accredited, independent laboratories. NABL (National Accreditation Board for Testing and Calibration Laboratories) is India's highest accreditation for testing laboratories, ensuring international comparability under ILAC-MRA. The results are legally defensible, regulatory-grade data.

#	Engine	Capacity	Site	Testing Lab (NABL)	Test Dates
A	Caterpillar C32 TA	1010 kVA	Pharma Co	Lata Envirotech Services	6 November 2025
B	Cummins	380 kVA	Pharma Co	Lata Envirotech Services	5 November 2025
C	Cummins	1500 kVA	Mfg Co	Ekdant Enviro Services (NABL + ISO 9001:2015)	14–18 October 2025
D	KOEL (250 kVA)	250 kVA	Mfg Co	Neetal Laboratories (NABL, ilac-MRA)	30 August 2025
E	KOEL 500 (2023 model)	500 kVA	Pharma Co	Neetal Laboratories (NABL, ilac-MRA)	23 December 2025

Engine A: Caterpillar C32 TA — 1010 kVA (Dr. Reddy's Laboratories, Vizag)

The Caterpillar C32 TA is a V-12, 32.1 litre, MEUI fuel system engine manufactured in 2006. Tested under actual pharmaceutical manufacturing loads at Dr. Reddy's Laboratories' Visakhapatnam facility.

Parameter	Baseline	6 Amps	10 Amps	13 Amps
PM (mg/Nm³)	68.42	63.16	52.40	**36.88 (−46%)**
NOx (ppm)	901	795.5	788.5	**589.5 (−35%)**
CO (%)	0.631	0.547	0.386	**0.370 (−41%)**
CO_2 (%)	9.15	7.85	7.65	**6.30 (−31%)**
THC (ppm)	609	569	328	**311 (−49%)**
O_2 (%)	11.76	13.08	13.09	**14.10 (+20%)**

Testing Lab: Lata Envirotech Services | Date: 6 November 2025 | Real industrial load at pharmaceutical manufacturing facility

Engine B: Cummins 380 kVA (Pharma Co.)

A 2018-model Cummins with electronic/common rail injection, tested simultaneously under the same factory load conditions as Engine A.

Parameter	Baseline	6 Amps	10 Amps	4 Amps (Optimal)
PM (mg/Nm³)	59.13	51.72 (−13%)	32.90 (−44%)	**20.84 (−65%)**
NOx (ppm)	274	232 (−15%)	202 (−26%)	**192 (−30%)**
SO_2 (ppm)	22	12 (−45%)	10 (−55%)	**8 (−64%)**
CO_2 (%)	5.3	4.5 (−15%)	4.4 (−17%)	**4.2 (−21%)**

Testing Lab: Lata Envirotech Services | Date: 5 November 2025

Engine C: Cummins 1500 kVA (Mfg Co.)

A large 2012-model Cummins with dual air intake, tested manufacturing facility under actual factory load. This is one of the most significant data sets — showing dramatic PM, SO_2, and NO_2 reductions at a large industrial DG scale.

Parameter	Baseline	300 ml/min	500 ml/min	700 ml/min
PM (mg/Nm³)	44.8	13.1 (−71%)	11.2 (−75%)	**8.9 (−80%)**
SO_2 (mg/Nm³)	31.4	12.9 (−59%)	10.3 (−67%)	**7.8 (−75%)**
NO_2 (mg/Nm³)	154.3	75.5 (−51%)	69.4 (−55%)	**62.9 (−59%)**
CO_2 (%)	5.6	5.1 (−9%)	4.8 (−14%)	**3.2 (−43%)**
O_2 (%)	12.0	13.8 (+15%)	14.6 (+22%)	**15.8 (+32%)**

Testing Lab: Ekdant Enviro Services Pvt. Ltd. (NABL + ISO 9001:2015) | Dates: 14–18 October 2025 | CO: Below Limit of Quantification at all conditions

Engine D: 250 kVA DG Set (Neetal Laboratories, Pune)

A 2012-model KOEL 250 kVA DG set tested at SGT's own facility with Neetal Laboratories (NABL, ilac-MRA) conducting independent sampling and testing to IS 11255 (Part 1) methodology.

Report No.	Load	HHO Flow	PM (mg/Nm³)	PM (Kg/kw.hr)	Temp °C	Change
RE/372	80%	None (Baseline)	413.0	0.403	432°C	Baseline
RE/373	80%	750 ml/min	89.9	0.083	449°C	**−78%**

RE/374	60%	None (Baseline)	1.3	0.0012	357°C	Baseline
RE/375	60%	500 ml/min	**Not Detected***	**Not Detected***	365°C	**< Detection**

*N.D. = Not Detected — below instrument detection limit. At 60% load with any HHO flow rate tested (160, 300, 500 ml/min), PM was entirely eliminated to below detection limits.

Engine E: KOEL 500 kVA — 2023 Model Year (Pharma Co.)

This is the most important test for regulatory and OEM purposes: a brand-new 2023-model KOEL engine, representing the latest CPCB 2 well-tuned platform. The test reveals SGT's critical insight about newer engines requiring lower, tighter dosing — a finding with profound implications for ARAI testing and CPCB recognition.

Parameter	Baseline	12–13 Amps	9–10 Amps	7–8 Amps (Optimal)
PM (mg/Nm³)	25.7	7.36 (−71%)	6.97 (−73%)	**6.1 (−76%)**
NOx (mg/Nm³)	810	687 (−15%)	653 (−19%)	**682 (−16%)**
CO_2 (%)	9.16	9.27 (+1.2%)*	7.95 (−13%)	**6.3 (−31%)**
CO (%)	0.587	0.57 (−3%)	0.41 (−30%)	**0.35 (−40%)**
HC (ppm)	409	397 (−3%)	277 (−32%)	**270 (−34%)**

Report No. NLES/25-26/12/ST/RE/1367 | Testing Lab: Neetal Laboratories (NABL, ilac-MRA) | Date: 23 December 2025

*At 12–13A the newer engine enters 'over-dosing zone'. This confirms SGT's fundamental finding: optimal dosing for 2023-model engines is significantly lower. SGT's adaptive algorithm automatically identifies and maintains the correct zone for each engine.

Fuel Efficiency Results & Peer-Reviewed Research

Fuel Efficiency — kWh/L Improvement

Beyond emission reduction, the primary economic benefit of GreenX™ is fuel savings. The key metric is kWh/L — kilowatt-hours of electricity generated per litre of diesel consumed. Higher kWh/L means less diesel burned per unit of power output.

Engine Platform	kWh/L Improvement Range	Best Load Condition
CPCB 2 platforms (older engines)	**8–15% improvement**	40–60% DG load (medium load)
CPCB 4 platforms (newer engines)	**4–13% improvement**	40–60% DG load (medium load)
High load (>75%)	Minimal incremental benefit	Baseline combustion near-optimal at high load

These efficiency improvements were measured using the kWh/L metric — actual electrical output per litre of diesel consumed — under real industrial factory load conditions. The simultaneous increase in exhaust O_2 and decrease in CO_2 across all platforms provides independent confirmation of reduced diesel consumption.

SGT's Peer-Reviewed Research Paper

'Controlled Hydrogen Fuel Assist via Intake Manifold Pre-Mixing as a Combustion Efficiency Enhancer in Multi-Cylinder Diesel Generator Engines: Experimental Investigation on 250–1500 kVA Genset Platforms'

Authors: Alok Kumar & Laxmikant Karande, SGT HydroEdge Private Limited

This research paper is the most comprehensive study of hydrogen fuel assist on industrial DG sets ever conducted in India. It presents:

- Full methodology using NABL-accredited lab testing at industrial customer sites — not a laboratory simulation
- Five DG platforms from 250 to 1500 kVA, covering different manufacturers (Caterpillar, Cummins, KOEL) and vintage years (2006 to 2023)
- 21 international peer-reviewed references from IJHE, Elsevier, MDPI, ScienceDirect, and Clean Energy journals
- The first formal definition of the three-zone model (Zone A/B/C) for CHFA — resolving contradictions in prior literature
- Detailed theoretical framework including P-θ diagrams, thermodynamic efficiency equations, and combustion kinetics analysis
- Explicit policy recommendations for CPCB and ARAI testing protocol development

International Research That Validates SGT's Findings

SGT's results are not isolated. 21 peer-reviewed international studies corroborate the fundamental science:

Study / Journal	Finding Corroborating SGT's Work
Sharma et al., IJHE, 2023 (Dry-cell HHO electrolyser on diesel engine)	Fuel consumption reductions of 5–12% depending on engine load and HHO flow rate
Liu et al., IJHE, 2020 (H_2 effects on combustion stability)	Up to ~10% H_2 energy fraction improves combustion stability; higher fractions cause instability — validates SGT's Zone B/C threshold
Kumar et al., Clean Energy, 2025 (H_2-assisted diesel engine)	Measurable improvements in brake thermal efficiency at low to moderate H_2 fractions; threshold beyond which gains plateau
Rimkus et al., IJHE, 2021 (H_2 in gasoline/LPG/diesel engines review)	H_2-diesel dual fuel shows promising efficiency at partial loads with diminishing returns at higher ratios — consistent with SGT's Zone model
Musmar et al., SSRN, 2022 (On-board HHO on diesel combustion)	On-board generated HHO reduces diesel consumption and emissions; NOx behaviour depends on dosing level
Ganesan et al., IJHE, 2025 (H_2 + water vapour injection)	Hydrogen serves as combustion enhancer while complementary strategies mitigate thermal NOx — confirming multi-mechanism operation
Mohamed et al., IJHE, 2010 (H_2-assisted diesel combustion review)	Foundational review identifying ignition delay reduction, premixed combustion enhancement as key mechanisms — directly matching SGT's empirical observations

The global academic consensus on HHO/hydrogen-assisted diesel combustion — drawn from researchers at University of Technology Baghdad, Wroclaw University, UNAM Mexico, NUST Pakistan, and multiple Indian, Turkish, Egyptian, Malaysian, and Ecuadorian institutions — confirms that SGT's results are scientifically sound, repeatable, and mechanistically well-understood.

Commercial Validation — Customer Evidence

The ultimate proof of any technology is not laboratory results — it is customers who have deployed it, measured results, and chosen to continue or expand. SGT has achieved this at two major industrial sites.

Customer 1: Pharma Co. — Two DG Sets Tested

Hyderabad Based, one of India's largest pharmaceutical companies (NSE/NYSE listed), hosted SGT GreenX™ installation and testing on two DG sets simultaneously at their manufacturing facility in November 2025:

- 1010 kVA Caterpillar C32 TA (2006 model) — Engine A in SGT's research paper
- 380 kVA Cummins (2018 model) — Engine B in SGT's research paper

Both tests were conducted under actual pharmaceutical manufacturing electrical loads — not simulated or bench conditions. NABL-accredited Lata Envirotech Services conducted independent stack emission measurements.

Results at Dr. Reddy's Laboratories:

Emission Parameter	1010 kVA Caterpillar (Best Result)	380 kVA Cummins (Best Result)
Particulate Matter (PM)	**46.1% reduction**	**64.8% reduction**
NOx	**34.6% reduction**	**29.9% reduction**
CO_2	**31.1% reduction**	**20.8% reduction**
SO_2	**44.6% reduction**	**63.6% reduction**
CO	**41.4% reduction**	**15.1% reduction**
THC	**48.9% reduction**	**9.3% reduction**

Customer 2: Mfg Co. — Repeat Orders Confirmed

This company is a leading manufacturer of turbochargers for automotive and industrial applications, hosted GreenX™ testing on their DG sets at their Chennai facility in October 2025:

- 1500 kVA Cummins (2012 model) — largest DG tested — Engine C in SGT's research paper
- 380 kVA Cummins (Engine at same site) — additional platform for cross-validation

Test Results at Turbo Energy:

Parameter	1500 kVA Cummins (Best Result)	380 kVA (Same Site, Best Result)
PM Reduction	**80.1% — Most dramatic PM reduction in entire study**	**75.0% reduction**

NO_2 Reduction	**59.2% reduction**	**71.9% reduction**
SO_2 Reduction	**75.2% reduction**	**62.9% reduction**
CO_2 Reduction	**42.9% reduction**	**28.3% reduction**

Commercial Outcome: Company has placed confirmed repeat orders for GreenX™ retrofit systems following the successful pilot. This is the single most powerful commercial signal — a customer who tested, verified results with independent NABL testing, and chose to purchase again without any subsidy or incentive pressure.

The Commercial Signal — Why Repeat Orders Matter

A first order could be influenced by relationship, pricing, or curiosity. A repeat order is a rational economic decision. When Turbo Energy placed repeat orders, they were communicating:

- The fuel savings are real and measurable in their actual operating accounts
- The system performed reliably through the initial deployment period
- The fuel and emission savings were verified against actual operating data
- Their finance and operations teams signed off on continued investment

This commercial signal, combined with the NABL-accredited 80.1% PM reduction at their 1500 kVA DG set, makes Turbo Energy one of the most credible reference customers in the Indian cleantech space for this technology category.

Scope 1 Emissions & the ESG Case for GreenX™

Every tonne of CO_2 that SGT eliminates is a tonne that a company did not have to account for, offset, or report as a liability. That is the fundamental promise of GreenX™ — and it begins at the asset level.

DGs Are a Scope 1 Liability That Most ESG Teams Overlook

When ESG teams map their organisation's Scope 1 carbon footprint, attention naturally falls on the largest and most visible combustion assets — industrial kilns, boilers, furnaces, and process heaters. These are the headline numbers in most sustainability reports.

Diesel generators, however, are a silent and persistent Scope 1 contributor that sits beneath the radar. Every litre of diesel burned in a DG set emits approximately 2.68 kg of CO_2, along with PM 2.5, NOx, CO, and hydrocarbons — all directly attributable to the organisation under GHG Protocol Scope 1 accounting. In a facility running multiple DG sets across backup and captive power duties, the cumulative annual carbon footprint is material — yet rarely targeted with the same rigour applied to process heating assets.

SGT's position is clear: "every asset that burns diesel is a Scope 1 asset." GreenX™ is the only validated retrofit solution that improves the combustion efficiency of that asset — reducing fuel consumption and therefore CO_2 at source — without any modification to the engine, process, or operations.

The Asset-Level Approach to Decarbonisation

SGT's philosophy is that decarbonisation must happen at the asset level — not just at the portfolio or offset level. Every engine that becomes more efficient is a permanent, verifiable, recurring reduction in Scope 1 emissions. Unlike carbon offsets or RECs, which are accounting instruments, GreenX™ creates a physical reduction in diesel combustion — fewer molecules of diesel burned per unit of power output, measurable in real time by NABL-accredited labs and continuously tracked by SGT's GreenVision™ IoT platform.

Asset Type	Scope 1 Emission Source	SGT Solution
Industrial Kilns	Fuel combustion — coal, gas, oil, biomass	SGT HHO technology reduces combustion inefficiency, cutting CO_2 and PM at source
Boilers (gas / oil / biomass)	Direct fuel burn; CO_2 and NOx emitted per unit of steam	Improved combustion completeness reduces fuel per tonne of steam produced
Diesel Generators	Every litre burned = 2.68 kg CO_2 + PM + NOx + CO	GreenX™ — up to 80% PM reduction, up to 43% CO_2 reduction. NABL-verified.
Marine Engines	Heavy fuel oil / diesel combustion at scale	GreenMarine™ — same combustion science, scaled for vessel engines

For ESG teams and Chief Sustainability Officers, the GreenX™ deployment on DG sets delivers something rare: an immediate, permanent, measurable reduction in Scope 1 emissions that requires no change to the asset, no change to the process, and no disruption to operations.

Why 'No Asset Change' Matters for ESG Deployment

One of the most common barriers to decarbonisation at the asset level is the complexity and risk of asset modification. Engine replacements, fuel switching, or process redesigns all carry engineering risk, capital expenditure, and operational disruption — and require board-level approvals that slow deployment.

GreenX™ removes every one of these barriers:

- Zero engine modification — the retrofit attaches to the air intake only. The DG set continues to operate exactly as before, with identical maintenance schedules, warranties, and OEM relationships.
- No change to fuel supply — GreenX™ uses water and the DG's own DC power supply. No new fuel infrastructure, no storage, no logistics.
- No operational disruption — installation takes approximately 4 hours. The DG set returns to full operation immediately.
- Immediate results from Day 1 — emission reductions begin with the first operating hour after installation, as confirmed by NABL-accredited before-and-after stack testing.
- Scalable across the entire DG fleet — every DG set from 25 kVA to 4,000 kVA, from any manufacturer, across any number of sites, can be retrofitted under the same framework.

For an ESG team facing a SEBI BRSR reporting deadline, an ESG audit, or a Net-Zero 2070 commitment, this means GreenX™ can be deployed quickly, reported immediately, and sustained indefinitely — without waiting for capital project approvals or engineering redesigns.

What GreenX™ Delivers for Your SEBI BRSR & ESG Reporting

ESG Requirement	What GreenX™ Delivers
Scope 1 GHG Reduction	Verified reduction in CO_2 per DG set — up to 43% per test. Directly reduces the Scope 1 number reported in BRSR and sustainability disclosures.
Air Quality / PM 2.5 Compliance	Up to 80.1% PM reduction (NABL-verified). Reduces particulate emissions from DG exhausts — a growing regulatory and community scrutiny point.
NOx & Local Pollutant Reduction	Up to 35% NOx reduction. Reduces site-level contribution to ground-level ozone and ambient air quality degradation.
Real-Time Emission Monitoring	GreenVision™ IoT platform provides continuous, asset-level Scope 1 data — ready for BRSR, internal ESG dashboards, and third-party audits.
Independent Verification	All results verified by NABL-accredited laboratories — the same standard used by CPCB-regulated testing. Data is audit-defensible.
No Greenwashing Risk	Physical combustion improvement — not offsets, not RECs, not accounting adjustments. Real emissions reduced at source. Every claim backed by NABL lab reports.

SGT's Guiding Philosophy: Sustainability with Profitability

At SGT, we believe that decarbonisation should never be a cost centre. Our founding philosophy — 'Sustainability with Profitability' — is not a tagline. It is the engineering and commercial standard to which every SGT product is held.

GreenX™ was designed from the ground up to deliver both simultaneously:

- **Sustainability: Meaningful, verified, permanent reduction in Scope 1 emissions across every DG set in operation — contributing directly to Net-Zero commitments, BRSR targets, and ESG ratings.**
- **Profitability: Fuel savings from improved combustion efficiency that translate directly into lower operating costs — making every deployment financially self-sustaining at SGT's competitive pricing, calibrated to the actual operating reality of DG sets in India today.**

India's DG sets today run 15 to 100 hours per month on average — a fraction of what they once did as grid reliability has improved. SGT understands this operating reality. Our commercial models and technology pricing are calibrated to this context, ensuring that even at reduced operating hours, the deployment delivers measurable ESG value that your sustainability team can report, your finance team can account for, and your board can stand behind.

The question for every ESG leader is not whether to decarbonise the DG fleet. The question is: how quickly can it be done, and with what level of verified evidence? GreenX™ answers both.

Regulatory Pathway — From NABL to CPCB Recognition

The Significance of ARAI Joint Testing

The current regulatory landscape for DG set emission retrofits in India has a gap: CPCB norms regulate new DG sets (up to CPCB IV+) but there is no formally recognized retrofit category for improving emissions from the millions of existing DG sets already in operation.

SGT is actively working to close this gap. ARAI's offer of joint testing creates a formal regulatory pathway:

Stage	Milestone	Status / Implication
1	**NABL-Accredited Field Testing**	COMPLETED. Five DG platforms, three NABL labs, data published in research paper. Regulatory-grade evidence established.
2	**ARAI Joint Testing**	IN PROGRESS. ARAI to conduct joint testing on GreenX™ under standardised ARAI test protocols. Unprecedented for a startup at SGT's stage.
3	**ARAI Test Report & Certification**	TARGET Q2-Q3 2026. ARAI test report will be Government of India recognised documentation of technology efficacy.
4	**CPCB Recognition**	TARGET 2026-27. ARAI test results submitted to CPCB for formal recognition of GreenX™ as an approved emission retrofit for legacy DG sets.
5	**Marine Extension**	PLANNED. Marine engines share the same diesel combustion architecture as DG sets. ARAI+CPCB recognition on DGs creates a direct pathway for marine validation.

Why This Creates a Durable Competitive Moat

Once ARAI and CPCB recognition is achieved, GreenX™ will be in a category occupied by no other hydrogen fuel assist technology in India. Competitors entering later would need to:

- Conduct their own multi-year testing programme from scratch
- Develop relationships with ARAI and CPCB independently
- Build a comparable field data set across DG manufacturers and sizes
- Overcome SGT's first-mover advantage in OEM partnerships (KOEL) and institutional endorsement (CII)

Regulatory recognition is not something money alone can buy quickly. It requires years of systematic testing, institutional trust, and policy engagement — all of which SGT is executing now as of March 2026.

Implementation & Engagement Models

Implementation Protocol

Stage	Phase	Activities	Output
1	**Baseline Audit (Weeks 1–2)**	DG set audit, load profile analysis, baseline stack emission test (NABL lab)	Baseline emissions report, optimal dosing design, commercial proposal
2	**Installation (Week 2–3)**	Retrofit to air intake, IoT sensor connection, GreenVision™ dashboard setup	System operational. No engine modification, no downtime beyond ~4 hours install.
3	**Optimisation (Weeks 4–8)**	Adaptive algorithm calibration to specific engine, load profile fine-tuning, post-installation NABL test	Calibrated system. Post-installation NABL test report documenting emission reductions.
4	**Ongoing Service**	Monthly GreenVision™ performance reports, scheduled maintenance, remote IoT monitoring	Continuous ESG/BRSR data. Audit-ready Scope 1 emission records. Real-time GreenVision™ dashboard.

Engagement Models

Model	Structure	Ideal For	Key Benefit
Enterprise DaaS (Buy + Service)	One-time equipment purchase + installation + monitoring + maintenance	Companies with CapEx budget seeking equipment ownership	Full ownership, low OpEx, SGT service support included
Subscription DaaS (Monthly Plan)	All-inclusive monthly fee: equipment, service, data, maintenance	Multi-DG fleet rollouts, long-term industrial contracts	Immediate Scope 1 emission reduction from Day 1. SGT retains equipment. Refundable security deposit.
Standard (Buy + AMC)	Equipment purchase + annual maintenance + GreenVision™ access	Single or small-volume deployments	Low-cost entry. Full monitoring and ESG reporting visibility.
Paid Pilot	Short-term pilot with full digital reporting. Flexible exit if targets unmet.	Technology evaluation before commitment to scale	Risk-free validation. One-time fee. End-to-end SGT support.

Investment Opportunity Summary

Saarthi GreenTech occupies a unique position in India's industrial decarbonisation landscape. The DG set segment represents a large, underserved, and immediately addressable market with no incumbent player offering a comparable validated solution.

Dimension	SGT's Position
Market Size	10+ million DG sets in India. Billions of litres of diesel per year. No recognised emission retrofit technology for legacy sets.
Technology Validation	5 platforms, 3 NABL labs, 1 peer-reviewed research paper, 21 international references. Most comprehensive dataset in India on this subject.
Regulatory Pathway	ARAI joint testing underway → CPCB recognition pathway. No competitor is at this stage.
OEM Endorsement	KOEL (Kirloskar Oil Engines Ltd) signed OEM pilot partnership. Access to India's largest DG distribution network.
Institutional Recognition	CII-ASCEND Top 10 Sustainability Company. Energy Leap Award 2025. Both with grants.
Scalability	Same core technology extends to marine engines (identical diesel combustion architecture). Global pilots in progress: India, Africa, Malaysia, Thailand.
IP Protection	Patent applied (India) for HHO generator design and adaptive dosing algorithm. Additional patents in process.
Decarbonisation Model	Decarbonisation-as-a-Service (DaaS) — recurring revenue model. GreenVision™ IoT SaaS platform for continuous monitoring and ESG reporting.

SGT is at an inflection point: from validated technology to scaled deployment. The ARAI testing, KOEL partnership, and repeat customer orders are the proof that the technology works. The opportunity now is to scale distribution, expand the DG customer base, and leverage the CPCB recognition pathway to create a regulatory moat that protects market leadership for years.

Contact & Next Steps

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To schedule a site visit, request NABL test reports, or initiate a paid pilot, please contact Alok Kumar directly.

"Decarbonisation made practical, measurable, and profitable."